# SSC MTS 2024 question paper with answers

**17 October 2024 · Shift 2**

77 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 13 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

## Numerical and Mathematical Ability 14 questions

**Q.3**

Five athletes are weighed consecutively, and after each measurement, their average weight is calculated. If the average weight increases by 1.5 kg each time, then how much heavier is the last player than the first player?

**(A)** 15.5 kg
**(B)** 12 kg
**(C)** 10 kg
**(D)** 18 kg

**Q.4**

It takes15 seconds for a 375-m long train to pass a pole. How long will it take to pass a 975-m long platform?

**(A)** 60 seconds
**(B)** 90 seconds
**(C)** 54 seconds
**(D)** 72 seconds

**Q.5**

What is the cost of a tadpole required to make a conical tent with slant height 15 m and height 9 m, when one square meter tadpole costs ₹147?

**(A)** Rs.49,896
**(B)** Rs.83,160
**(C)** Rs.62,370
**(D)** Rs.67,842

**Q.6**

A thief steels a bike at 10:30 am and drives it at 50 km/hr. But the thief is found after half an hour. The bike owner started to chase him by another bike at 60 km/hr. The time at which the thief is caught is:

**(A)** 12: 30 pm
**(B)** 1: 00 pm
**(C)** 2: 30 pm
**(D)** 1: 30 pm

**Q.8**

The greatest number of four digits which is divisible by each one of the numbers 15, 25, 40 and 75 is:

**(A)** 9500
**(B)** 9800
**(C)** 9400
**(D)** 9600

**Q.9**

Rakesh obtained a simple interest of ₹3,195 on a certain principal in 6 years at the rate of 15% per annum. What amount of simple interest would he get if he invested ₹1,250 more than the previous principal for 3 years at the same rate of interest?

**(A)** ₹2,255
**(B)** ₹2,050
**(C)** ₹2,160
**(D)** ₹2,400

**Q.12**

A conical cavity is drilled in a solid circular cylinder of 7 cm height and 48 cm base diameter. The height and base diameter of the cone are the same as those of the cylinder. What is the total surface area of the remaining solid?

**(A)** 1420π $cm^2$

**(B)** 1654π $cm^2$

**(C)** 1512π $cm^2$

**(D)** 1076π $cm^2$

**Q.13**

When the price of a laptop was reduced by 15%, its sales increased by 10%. What was the net effect on the sale?

**(A)** 3.5% increase

**(B)** 3.5% decrease

**(C)** 6.5% decrease

**(D)** 6.5% increase

**Q.15**

Find the simple interest (in ₹) on ₹2000 as a sum borrowed at 7% per year rate of interest for 6 years.

**(A)** 840

**(B)** 780

**(C)** 800

**(D)** 880

**Q.16**

The average age of a class of 39 students is 20 years. If the age of the teacher is included, then the average increases by 6 months. Find the age of the teacher (in years).

**(A)** 30

**(B)** 40

**(C)** 50

**(D)** 45

**Q.17**

In what proportion must a man mix milk worth ₹3.3/litre with milk worth ₹1.8/litre so that the mixture may be worth ₹2.4/litre?

**(A)** 4 : 5

**(B)** 2 : 3

**(C)** 3 : 2

**(D)** 1 : 3

**Q.18**

Find the simple interest (in ₹) on ₹2500 as a sum borrowed at 5% per year rate of interest for 3 years.

**(A)** 335

**(B)** 415

**(C)** 375

**(D)** 315

**Q.19**

The average weight of a family of five members (in kg) whose weights are 40 kg, 49 kg, 56 kg, 71 kg and 38 kg is:

**(A)** 52.8

**(B)** 49.8

**(C)** 51.8

**(D)** 50.8

**Q.20**

Ramesh purchased a piano, marked at ₹18,000 at a successive discounts of 12% and 15%, respectively. Ramesh gave ₹536 as transportation charge and sold it at ₹21,000. What is his gain percentage?

**(A)** 35%

**(B)** 48%

**(C)** 50%

**(D)** 40%

## Reasoning Ability and Problem Solving 16 questions

**Q.1**

17 is related to 119 following a certain logic. Following the same logic, 31 is related to 217. To which of the following is 32 related, following the same logic? (NOTE : Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding / subtracting / multiplying, etc., to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

**(A)** 226
**(B)** 228
**(C)** 230
**(D)** 224

**Q.4**

What will come in the place of '?' in the following equation, if '+' and '–' are interchanged and '×' and '÷' are interchanged? $14 \div 6 + 9 \times 3 - 2 = ?$

**(A)** 76
**(B)** 75
**(C)** 81
**(D)** 83

**Q.5**

A, B, C, D, E, F and G are sitting around a circular table facing the centre (but not necessarily in the same order). Only one person sits between D and A when counted from the right of D. Only one person sits between A and B. Only one person sits between D and G. Only two people sit between E and B, when counted from the right of E. Only one person sits between F and G. Who sits third to the right of C ?

**(A)** D
**(B)** A
**(C)** F
**(D)** B

**Q.6**

Select the set in which the numbers are related in the same way as are the numbers of the following sets. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into their constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) (37, 58, 100) (35, 70, 110)

**(A)** (85, 75, 65)
**(B)** (74, 33, 12)
**(C)** (43, 44, 132)
**(D)** (8, 13, 26)

**Q.7**

Read the given statements and conclusions carefully. You have to take the given statements to be true even if they seem to be at variance from commonly known facts. You have to decide which conclusion/s logically follow/s from the given statements. Statements: All boxes are glasses. All glasses are cups. Some cups are bottles. Conclusions: (I): All boxes are cups. (II): Some glasses are bottles.

**(A)** Both conclusions I and II follow.
**(B)** Neither conclusion I nor II follows.
**(C)** Only conclusion II follows.
**(D)** Only conclusion I follows.

**Q.8**

IOZU is related to HNYT in a certain way based on the English alphabetical order. In the same way, PDYC is related to OCXB. To which of the following is EJSW related, following the same logic?

**(A)** DIRW
**(B)** EIRV
**(C)** DIRV
**(D)** DJSV

**Q.9**

In a certain code,'curtain door window' is coded as 'la ph ba','lock window key' is coded as 'ep zm la','key curtain handle' is coded as 'fd ba zm'. What is the code for 'curtain' in that language?(Note: All the codes are two-letter coded only.)

**(A)** zm
**(B)** la
**(C)** ba
**(D)** ep

**Q.10**

What will come in place of the question mark (?) in the following equation, if '+' and '–' are interchanged and '×' and '÷' are interchanged? 66 × 6 – 15 ÷ 3 + 9 = ?

**(A)** 49
**(B)** 47
**(C)** 48
**(D)** 46

**Q.11**

A, D, K, N, Y and Z are sitting around a circular table facing the centre (but not necessarily in the same order). Z is sitting third to the right of D. K is not the immediate neighbour of Z. A is sitting second to the left of Z. N is not the immediate neighbour of A. Who is the immediate neighbour of both between A and K?

**(A)** D
**(B)** Y
**(C)** N
**(D)** Z

**Q.13**

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: Some books are wood. All wood is fire. Conclusions: (I) Some fire is wood. (II) All books are fire.

**(A)** Both conclusions (I) and (II) follow
**(B)** Only conclusion (II) follows
**(C)** Neither conclusion (I) nor (II) follows
**(D)** Only conclusion (I) follows

**Q.14**

What should come in place of the question mark (?) in the given series? 219 222 227 234 ? 254

**(A)** 243
**(B)** 242
**(C)** 244
**(D)** 241

**Q.15**

What should come in place of the question mark (?) in the given series based on the English alphabetical order? GJD ,IHH, KFL, MDP, ?

**(A)** PBT
**(B)** PAT
**(C)** OBS
**(D)** OBT

**Q.16**

In a certain code language, 'complete the task' is written as 'je tc xo' and 'work is complete' is coded as 'xo ig lx'. How is 'complete' coded in the given language?

**(A)** tc
**(B)** je
**(C)** lx
**(D)** xo

**Q.17**

OT11 is related to PU15 in a certain way. In the same way, WB21 is related to XC25. To which of the following is DI31 related, following the same logic?

**(A)** DJ35
**(B)** EJ35
**(C)** EF53
**(D)** EJ53

**Q.18**

In a certain code language, ‘this is book’ is written as ‘lo mo ro’ and ‘this is me’ is written as ‘mo lo po’. How is ‘me’ written in the given code language?

**(A)** lo
**(B)** ro
**(C)** mo
**(D)** po

**Q.20**

In a certain code language, ‘FITCH’ is coded as ‘47601’ and ‘DITCH’ is coded as ‘70619’. What is the code for ‘D’ in the given code language?

**(A)** 1
**(B)** 7
**(C)** 9
**(D)** 0

## General Awareness 23 questions

**Q.2**

‘Bayalata’ is a famous folk dance of which of the following Indian states?

**(A)** Himachal Pradesh
**(B)** Bihar
**(C)** Punjab
**(D)** Karnataka

**Q.3**

In which year was the first session of the Indian National Congress held?

**(A)** 1882
**(B)** 1886
**(C)** 1885
**(D)** 1887

**Q.5**

In a laterite soil profile, which of the following elements is present in an excessive amount?

**(A)** feldspar
**(B)** Nitrogen
**(C)** Lime
**(D)** Iron

**Q.6**

What is the chemical reaction that occurs when sugar is heated and turns into a caramel syrup?

**(A)** Precipitation
**(B)** Neutralisation
**(C)** Caramelisation
**(D)** Maillard reaction

**Q.7**

Who among the following is known as the 'Mozart of Madras'?

**(A)** Shankar Mahadevan
**(B)** MM Kreem
**(C)** AR Rahman
**(D)** Ilaiyaraaja

**Q.8**

Devendra Fadnavis became the _____ of the state of Maharashtra on 30th June 2022.

**(A)** Chief Justice of High Court
**(B)** Deputy Chief Minister
**(C)** Governor
**(D)** Chief Minister

**Q.9**

Which of the following States/Union Territories is NOT a part of Indus River Basin?

**(A)** Himachal Pradesh
**(B)** Ladakh
**(C)** Jammu and Kashmir
**(D)** Sikkim

**Q.10**
The Fundamental Rights Sub-Committee of the Constituent Assembly was headed by _____.

**(A)** Harnam Singh
**(B)** Alladi Krushnaswami Ayyar
**(C)** JB Kripalani
**(D)** Jawaharlal Nehru

**Q.11**
In which of the following months is the Konark Dance festival organised in Odisha?

**(A)** March
**(B)** June
**(C)** January
**(D)** December

**Q.12**
The Hathigumpha Inscription was inscribed in which of the following historical regions of India?

**(A)** Kalinga
**(B)** Vaishali
**(C)** Taxila
**(D)** Suvarnagiri

**Q.13**
The transportation mechanism in cells by which food or other useful molecules from the outside are carried inside the cell is known as _____.

**(A)** Endocytosis
**(B)** Osmosis
**(C)** Exocytosis
**(D)** Fusion

**Q.14**
Where does the cash grants in aid from foreign countries are classified?

**(A)** Non-tax revenue
**(B)** Capital receipt
**(C)** Tax revenue
**(D)** Revenue expenditure

**Q.15**
Sachin Tendulkar scored his 100th international century against which country?

**(A)** Australia
**(B)** England
**(C)** Bangladesh
**(D)** Pakistan

**Q.16**
Which of the following para athletes is associated with the javelin throw event?

**(A)** Sharad Kumar
**(B)** Devendra Jhajharia
**(C)** Pramod Bhagat
**(D)** Karamjyoti Dalal

**Q.17**
When was ‘Nai Chetna-Pahal Badlav Ki’ campaign launched?

**(A)** November 2022
**(B)** December 2022
**(C)** January 2023
**(D)** March 2023

**Q.18**
What is the full form of INSAT?

**(A)** Indian Navigational Satellite System
**(B)** International Navigation Satellite System
**(C)** Indian National Standard System
**(D)** Indian National Satellite System

**Q.19**
Under the influence of the Greeks, which of the following art styles flourished in India?

**(A)** Gandhar
**(B)** Mathura
**(C)** Amravati
**(D)** Kandhar

**Q.20**
National Urban Livelihood Mission (NULM) was launched in which of the following Five Year plans?

**(A)** Tenth Five Year Plan
**(B)** Sixth Five Year Plan
**(C)** Eighth Five Year Plan
**(D)** Twelfth Five Year Plan

**Q.21**
According to Census 2011 of India, which Union Territory has the second highest percentage of urban population?

**(A)** Daman and Diu
**(B)** Delhi
**(C)** Chandigarh
**(D)** Puducherry

**Q.22**
'Unbreakable' is an autobiography of _____ published in 2013.

**(A)** Abhinav Bindra
**(B)** MC Mary Kom
**(C)** Milkha Singh
**(D)** Geeta Phogat

**Q.23**
Which of the following events took place in the year 1928?

**(A)** Kakori Incident
**(B)** Bardoli Satyagraha
**(C)** Kheda Satyagraha
**(D)** Dandi March

**Q.24**
As per the treaty of Purandar, how many forts were surrendered by Shivaji to Mughals?

**(A)** 23
**(B)** 18
**(C)** 11
**(D)** 35

**Q.25**
The number of Rajya Sabha members to be elected from each State of India has been fixed by the _____ Schedule of the Constitution.

**(A)** Sixth
**(B)** Eighth
**(C)** Third
**(D)** Fourth

## English Language and Comprehension 24 questions

**Q.1**
Select the most appropriate option to fill in the blank. In order to maintain good governance, every government servant must be held _____ for his/her decisions.

**(A)** irresponsible
**(B)** accountable
**(C)** unaccountable
**(D)** flexible

**Q.2**
Select the most appropriate ANTONYM that can substitute the underlined segment in the given sentence. The team was gradually realising that the social gatherings were “draining”.

**(A)** absorbing
**(B)** flippant
**(C)** unfruitful
**(D)** frivolous

**Q.4**
Select the INCORRECTLY spelt word in the given sentence. Many biologists have stated the fact that most of the herbs are perinnial and have healing qualities with added flavour.

**(A)** Biologists
**(B)** Healing
**(C)** Perinnial
**(D)** Qualities

**Q.5**

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. In addition with / the existing crisis in economy, / the rise of hunger index / is another cause of worry.

**(A)** In addition with
**(B)** the rise of hunger index
**(C)** the existing crisis in economy
**(D)** is another cause of worry

**Q.6**

Select the most appropriate option to fill in the blank. The inclusion of Durga Puja in the UNESCO's list of Intangible Cultural Heritage was only possible due to the arduous _____ of all stakeholders.

**(A)** efforts
**(B)** passivity
**(C)** affects
**(D)** ignorance

**Q.7**

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. My sister Mona / never drinks coffee; / she prefer tea / in the morning.

**(A)** in the morning
**(B)** she prefer tea
**(C)** never drinks coffee;
**(D)** My sister Mona

**Q.8**

Select the INCORRECTLY spelt word.

**(A)** Machinary
**(B)** Motto
**(C)** Idol
**(D)** Hymn

**Q.9**

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. Most probably, / he come / to office / next Tuesday.

**(A)** next Tuesday
**(B)** most probably
**(C)** to office
**(D)** he come

**Q.10**

Select the option that will improve the underlined part of the given sentence. The bishop sadly declared that the poor gentleman who was dead was “quieter” of the community.

**(A)** most quiet
**(B)** the quietest
**(C)** quiet
**(D)** the quieter

**Q.11**

Select the INCORRECTLY spelt word in the given sentence. The neighbour came in disgoise and attacked her children.

**(A)** attacked
**(B)** disgoise
**(C)** children
**(D)** neighbour

**Q.12**

Select the most appropriate option that can substitute the underlined word in the given sentence. Reduced price of cooking gas brought “happy” to the face of every citizen.

**(A)** happiest
**(B)** happily
**(C)** happier
**(D)** happiness

**Q.13**

Select the most appropriate option that can substitute the underlined segment in the given sentence. If there is no need to substitute it, select 'No substitution required'. Rishi Sunak, the Prime Minister of the United Kingdom and Leader of the Conservative Party, “was born into” Southampton to parents of Indian descent who immigrated to Britain from East Africa in the 1960s.

**(A)** was born in
**(B)** was born on
**(C)** No substitution required
**(D)** was born with

**Q.14**

Select the option that will improve the underlined part of the given sentence. The moment James saw Regena, he fell “head on to his heels” in love with her.

**(A)** heels of the head
**(B)** head over heels
**(C)** head and heels
**(D)** heads in the heels

**Q.15**

Select the most appropriate meaning of the given idiom. Beat around the bush

**(A)** To speak rashly
**(B)** To avoid getting to the point of an issue
**(C)** To speak convincingly
**(D)** To speak abusively

**Q.16**

Select the most appropriate ANTONYM of the bracketed word to fill in the blank. The rainbow arched gracefully across the sky, painting it in _____ (lifeless) hues.

**(A)** muted
**(B)** vibrant
**(C)** spiritual
**(D)** ashless

**Q.17**

Select the most appropriate ANTONYM of the given word. Pacify

**(A)** Restore
**(B)** Negate
**(C)** Enrage
**(D)** Amuse

**Q.18**

Read the following sentence and select its future tense form from the following options. The food company promoted newly launched products at affordable prices.

**(A)** The food company is promoting newly launched products at affordable prices.
**(B)** The food company will be promoting newly launched products at affordable prices.
**(C)** The food company promotes newly launched products at affordable prices.
**(D)** The food company had promoted newly launched products at affordable prices.

**Q.19**

Select the most appropriate adverb to fill in the blank. The explorers trekked _____ through the dense jungle.

**(A)** bravely
**(B)** before
**(C)** rarely
**(D)** ever

**Q.20**

Select the most appropriate ANTONYM of the given word. Delay

**(A)** Hasten
**(B)** Lag
**(C)** Intervene
**(D)** Wait

**Q.21**

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. We're a society fixated on numbers. So, it's no (1)_____ we use measurements and equations to score our weight. The most (2)_____ is BMI, a measure of our body weight to height ratio. BMI (3)_____ bodies as underweight, normal or healthy weight, overweight or obese and can be a useful tool for weight and health (4)_____. But it shouldn't be used as the single (5)_____ of what it means to be a healthy weight when we set our weight loss goals.

Select the most appropriate option to fill in blank number 1.

**(A)** surprise
**(B)** vibrance
**(C)** persistence
**(D)** resolution

**Q.22**

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. We're a society fixated on numbers. So, it's no (1)_____ we use measurements and equations to score our weight. The most (2)_____ is BMI, a measure of our body weight to height ratio. BMI (3)_____ bodies as underweight, normal or healthy weight, overweight or obese and can be a useful tool for weight and health (4)_____. But it shouldn't be used as the single (5)_____ of what it means to be a healthy weight when we set our weight loss goals.

Select the most appropriate option to fill in blank number 2.

**(A)** uncharted
**(B)** indistinct
**(C)** imprecise
**(D)** popular

**Q.23**

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. We're a society fixated on numbers. So, it's no (1)_____ we use measurements and equations to score our weight. The most (2)_____ is BMI, a measure of our body weight to height ratio. BMI (3)_____ bodies as underweight, normal or healthy weight, overweight or obese and can be a useful tool for weight and health (4)_____. But it shouldn't be used as the single (5)_____ of what it means to be a healthy weight when we set our weight loss goals.

Select the most appropriate option to fill in blank number 3.

**(A)** speculates
**(B)** meanders
**(C)** dissuades
**(D)** classifies

**Q.24**

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. We're a society fixated on numbers. So, it's no (1)_____ we use measurements and equations to score our weight. The most (2)_____ is BMI, a measure of our body weight to height ratio. BMI (3)_____ bodies as underweight, normal or healthy weight, overweight or obese and can be a useful tool for weight and health (4)_____. But it shouldn't be used as the single (5)_____ of what it means to be a healthy weight when we set our weight loss goals.

Select the most appropriate option to fill in blank number 4.

**(A)** screening
**(B)** landing
**(C)** hastening
**(D)** planting

**Q.25**

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. We're a society fixated on numbers. So, it's no (1)_____ we use measurements and equations to score our weight. The most (2)_____ is BMI, a measure of our body weight to height ratio. BMI (3)_____ bodies as underweight, normal or healthy weight, overweight or obese and can be a useful tool for weight and health (4)_____. But it shouldn't be used as the single (5)_____ of what it means to be a healthy weight when we set our weight loss goals.

Select the most appropriate option to fill in blank number 5.

**(A)** measure

**(B)** serendipity

**(C)** evanesce

**(D)** wobble

## Answer key

### Numerical and Mathematical Ability

| | | | | | |
|---|---|---|---|---|---|
| Q.3 **(B)** | Q.4 **(C)** | Q.5 **(B)** | Q.6 **(D)** | Q.8 **(D)** | Q.9 **(C)** |
| Q.12 **(C)** | Q.13 **(C)** | Q.15 **(A)** | Q.16 **(B)** | Q.17 **(B)** | Q.18 **(C)** |
| Q.19 **(D)** | Q.20 **(C)** | | | | |

### Reasoning Ability and Problem Solving

| | | | | | |
|---|---|---|---|---|---|
| Q.1 **(D)** | Q.4 **(D)** | Q.5 **(B)** | Q.6 **(D)** | Q.7 **(D)** | Q.8 **(C)** |
| Q.9 **(C)** | Q.10 **(B)** | Q.11 **(A)** | Q.13 **(D)** | Q.14 **(A)** | Q.15 **(D)** |
| Q.16 **(D)** | Q.17 **(B)** | Q.18 **(D)** | Q.20 **(C)** | | |

### General Awareness

| | | | | | |
|---|---|---|---|---|---|
| Q.2 **(D)** | Q.3 **(C)** | Q.5 **(D)** | Q.6 **(C)** | Q.7 **(C)** | Q.8 **(B)** |
| Q.9 **(D)** | Q.10 **(C)** | Q.11 **(D)** | Q.12 **(A)** | Q.13 **(A)** | Q.14 **(A)** |
| Q.15 **(C)** | Q.16 **(B)** | Q.17 **(A)** | Q.18 **(D)** | Q.19 **(A)** | Q.20 **(D)** |
| Q.21 **(C)** | Q.22 **(B)** | Q.23 **(B)** | Q.24 **(A)** | Q.25 **(D)** | |

### English Language and Comprehension

| | | | | | |
|---|---|---|---|---|---|
| Q.1 **(B)** | Q.2 **(A)** | Q.4 **(C)** | Q.5 **(A)** | Q.6 **(A)** | Q.7 **(B)** |
| Q.8 **(A)** | Q.9 **(D)** | Q.10 **(B)** | Q.11 **(B)** | Q.12 **(D)** | Q.13 **(A)** |
| Q.14 **(B)** | Q.15 **(B)** | Q.16 **(B)** | Q.17 **(C)** | Q.18 **(B)** | Q.19 **(A)** |
| Q.20 **(A)** | Q.21 **(A)** | Q.22 **(D)** | Q.23 **(D)** | Q.24 **(A)** | Q.25 **(A)** |

## Explanations

### Numerical and Mathematical Ability

**Q.3 · Answer (B) 12 kg**

If the first athlete weighs w, the averages are w, w + 1.5, w + 3, w + 4.5 and w + 6, so the totals are w, 2w + 3, 3w + 9, 4w + 18 and 5w + 30. The last athlete weighs (5w + 30) – (4w + 18) = w + 12: 12 kg more than the first.

**Q.4 · Answer (C) 54 seconds**

Speed = 375/15 = 25 m/s. To pass the platform the train covers 375 + 975 = 1,350 m: 1,350/25 = 54 seconds.

**Q.5 · Answer (B) Rs.83,160**

The radius is $\sqrt{(15^2 - 9^2)}$ = 12 m, so the canvas (the 'tadpole', i.e. tarpaulin) covers πrl = (22/7) × 12 × 15 = 3,960/7 m². At ₹147 a square metre: 3,960/7 × 147 = ₹83,160.

**Q.6 · Answer (D) 1: 30 pm**

By 11:00 am, when the chase begins, the thief is 50 × ½ = 25 km ahead. The owner gains 60 – 50 = 10 km/h, so he takes 25/10 = 2.5 hours: 1:30 pm.

**Q.8 · Answer (D) 9600**

LCM of 15, 25, 40 and 75 = $2^3 \times 3 \times 5^2$ = 600. The greatest four-digit multiple of 600 is 600 × 16 = 9600.

**Q.9 · Answer (C) ₹2,160**

Principal = 3,195 × 100/(15 × 6) = ₹3,550. The new principal is 3,550 + 1,250 = ₹4,800, and its interest for 3 years is 4,800 × 15 × 3/100 = ₹2,160.

**Q.12 · Answer (C) 1512π cm²**

Radius 24 cm; the cavity's slant height is $\sqrt{(24^2 + 7^2)}$ = 25 cm. The remaining solid's surface is the cylinder's curved surface, its base and the cone's curved surface: 2π × 24 × 7 + π × $24^2$ + π × 24 × 25 = 336π + 576π + 600π = 1512π cm².

**Q.13 · Answer (C) 6.5% decrease**

Sales value = price × quantity: 0.85 × 1.10 = 0.935 of the old value, a 6.5% decrease.

**Q.15 · Answer (A) 840**

SI = 2000 × 7 × 6/100 = ₹840.

**Q.16 · Answer (B) 40**

The 39 students' ages total 39 × 20 = 780 years. With the teacher, 40 people average 20.5 years, a total of 820, so the teacher is 820 – 780 = 40.

**Q.17 · Answer (B) 2 : 3**

By alligation, milk at ₹3.3 and at ₹1.8 is mixed in the ratio (2.4 – 1.8) : (3.3 – 2.4) = 0.6 : 0.9 = 2 : 3.

**Q.18 · Answer (C) 375**

SI = 2500 × 5 × 3/100 = ₹375.

**Q.19 · Answer (D) 50.8**

(40 + 49 + 56 + 71 + 38)/5 = 254/5 = 50.8 kg.

**Q.20 · Answer (C) 50%**

Price paid = 18,000 × 0.88 × 0.85 = ₹13,464; with ₹536 transport the cost is ₹14,000. Gain = 21,000 – 14,000 = ₹7,000, which is 50% of 14,000.

### Reasoning Ability and Problem Solving

**Q.1 · Answer (D) 224**

Each number is multiplied by 7: 17 × 7 = 119 and 31 × 7 = 217. Likewise 32 × 7 = 224.

**Q.4 · Answer (D) 83**

After interchanging the signs: 14 × 6 – 9 ÷ 3 + 2 = 84 – 3 + 2 = 83.

**Q.5 · Answer (B) A**
Seated clockwise (each person's left, facing the centre): D, C, G, B, F, A, E, which meets every condition. Going from C to its right (anticlockwise): D, E, A, so A is third to the right of C.

**Q.6 · Answer (D) (8, 13, 26)**
The third number is the sum of the first two plus 5: 37 + 58 + 5 = 100 and 35 + 70 + 5 = 110. Likewise 8 + 13 + 5 = 26.

**Q.7 · Answer (D) Only conclusion I follows.**
All boxes are glasses and all glasses are cups, so all boxes are cups (I). The cups that are bottles need not be glasses, so II does not follow.

**Q.8 · Answer (C) DIRV**
Each letter moves 1 place back: IOZU → HNYT and PDYC → OCXB. So EJSW → DIRV.

**Q.9 · Answer (C) ba**
'window' is in the first two phrases and 'la' in both their codes, so window = la; 'key' is in the last two and 'zm' in both their codes, so key = zm. 'curtain' is in the first and the last, and the code they share is 'ba'.

**Q.10 · Answer (B) 47**
After interchanging the signs: 66 ÷ 6 + 15 × 3 – 9 = 11 + 45 – 9 = 47.

**Q.11 · Answer (A) D**
Seated clockwise (each person's left, facing the centre): D, K, N, Z, Y, A. Z is opposite D, A is second to Z's left, and K and N then take the only seats that keep them away from Z and A. D sits between A and K.

**Q.13 · Answer (D) Only conclusion (I) follows**
Some books are wood and all wood is fire, so some fire is wood (I). The books that are not wood need not be fire, so II does not follow.

**Q.14 · Answer (A) 243**
The differences are 3, 5, 7, 9, 11: 234 + 9 = 243, and 243 + 11 = 254.

**Q.15 · Answer (D) OBT**
First letters move 2 forward (G, I, K, M, so O), second letters 2 back (J, H, F, D, so B) and third letters 4 forward (D, H, L, P, so T): OBT.

**Q.16 · Answer (D) xo**
'complete' is the only word in both sentences and 'xo' the only code in both, so complete = xo.

**Q.17 · Answer (B) EJ35**
Each letter moves 1 place forward and the number rises by 4: OT11 → PU15 and WB21 → XC25. So DI31 → EJ35.

**Q.18 · Answer (D) po**
'this' and 'is' are in both sentences, as are 'lo' and 'mo' in both codes. The word left in the second sentence is 'me' and the code left is 'po', so me = po.

**Q.20 · Answer (C) 9**
I, T, C and H are in both words, as are 7, 6, 0 and 1 in both codes. The letter left in DITCH is D and the digit left in its code is 9, so D = 9.

### General Awareness

**Q.2 · Answer (D) Karnataka**
Bayalata, literally 'open-field play', is a folk dance-drama of Karnataka performed in the open air.

**Q.3 · Answer (C) 1885**
The first session of the Indian National Congress met in Bombay in December 1885, presided over by W.C. Bonnerjee.

**Q.5 · Answer (D) Iron**
Laterite soils form under heavy rain and high heat, which leach away lime and silica; what remains is rich in iron oxide and aluminium compounds.

**Q.6 · Answer (C) Caramelisation**
Heated on its own, sugar breaks down into the brown, flavourful compounds of caramel: caramelisation. The Maillard reaction needs amino acids (proteins) as well as sugars.

**Q.7 · Answer (C) AR Rahman**
The composer A.R. Rahman, born in Madras (Chennai), is called the 'Mozart of Madras'.

**Q.8 · Answer (B) Deputy Chief Minister**
Devendra Fadnavis was sworn in as Deputy Chief Minister of Maharashtra on 30 June 2022, when Eknath Shinde became Chief Minister.

**Q.9 · Answer (D) Sikkim**
Sikkim is drained by the Teesta, part of the Brahmaputra system. Ladakh, Jammu and Kashmir and Himachal Pradesh are drained mainly by the Indus and its tributaries.

**Q.10 · Answer (C) JB Kripalani**
The Constituent Assembly's sub-committee on Fundamental Rights was chaired by J.B. Kripalani, under the Advisory Committee headed by Sardar Patel.

**Q.11 · Answer (D) December**
The Konark Dance Festival is held every year from 1 to 5 December at the open-air auditorium near the Sun Temple.

**Q.12 · Answer (A) Kalinga**
The Hathigumpha inscription, in the Udayagiri caves near Bhubaneswar, records the reign of Kharavela, king of Kalinga.

**Q.13 · Answer (A) Endocytosis**
In endocytosis the cell membrane folds round material outside the cell and takes it in; exocytosis is the reverse, and osmosis is the movement of water across a membrane.

**Q.14 · Answer (A) Non-tax revenue**
In the government budget, cash grants-in-aid from foreign countries and international organisations are counted as non-tax revenue, along with interest, dividends, profits and fees.

**Q.15 · Answer (C) Bangladesh**
Sachin Tendulkar's 100th international century came against Bangladesh at Mirpur, in the 2012 Asia Cup.

**Q.16 · Answer (B) Devendra Jhajharia**
Devendra Jhajharia won Paralympic gold in the javelin throw in 2004 and 2016. Sharad Kumar is a high jumper and Pramod Bhagat a badminton player.

**Q.17 · Answer (A) November 2022**
The Ministry of Rural Development (DAY-NRLM) launched 'Nai Chetna', a national campaign against gender-based violence, on 25 November 2022.

**Q.18 · Answer (D) Indian National Satellite System**
INSAT is the Indian National Satellite System, ISRO's series of geostationary satellites for communication, broadcasting and weather services.

**Q.19 · Answer (A) Gandhar**
Gandhara art, which grew up in the north-west under Greek (Hellenistic) influence, shows the Buddha in a Greco-Roman style. The Mathura and Amaravati schools were largely indigenous.

**Q.20 · Answer (D) Twelfth Five Year Plan**
The National Urban Livelihoods Mission was launched in September 2013, during the Twelfth Five Year Plan (2012-17), replacing the Swarna Jayanti Shahari Rozgar Yojana.

**Q.21 · Answer (C) Chandigarh**
In the 2011 census Delhi (97.5%) had the highest share of urban population among the Union Territories, followed by Chandigarh (97.25%).

**Q.22 · Answer (B) MC Mary Kom**
'Unbreakable' (2013) is the autobiography of the boxer M.C. Mary Kom.

**Q.23 · Answer (B) Bardoli Satyagraha**
The Bardoli Satyagraha, led by Vallabhbhai Patel against a rise in land revenue, took place in 1928. The Kakori incident was in 1925, the Kheda Satyagraha in 1918 and the Dandi March in 1930.

**Q.24 · Answer (A) 23**
By the Treaty of Purandar (1665) with Raja Jai Singh, Shivaji surrendered 23 forts to the Mughals and kept 12.

**Q.25 · Answer (D) Fourth**
The Fourth Schedule allocates the seats in the Rajya Sabha to the States and Union Territories.

### English Language and Comprehension

**Q.1 · Answer (B) accountable**
Good governance needs servants who answer for their decisions: 'held accountable'.

**Q.2 · Answer (A) absorbing**
'Draining' means tiring and wearing; its opposite here is 'absorbing', holding one's interest.

**Q.4 · Answer (C) Perinnial**
The correct spelling is 'perennial'.

**Q.5 · Answer (A) In addition with**
The phrase is 'in addition to': 'In addition to the existing crisis in economy'.

**Q.6 · Answer (A) efforts**
The inclusion was possible due to the 'arduous efforts', the hard work, of all stakeholders; 'affects' is a verb.

**Q.7 · Answer (B) she prefer tea**
A third-person singular subject takes 'prefers': 'she prefers tea in the morning'.

**Q.8 · Answer (A) Machinary**
The correct spelling is 'machinery'.

**Q.9 · Answer (D) he come**
'Next Tuesday' is in the future, so 'he will come'; 'he come' also lacks the singular verb form.

**Q.10 · Answer (B) the quietest**
'Of the community' compares him with everyone in it, so the superlative: 'the quietest of the community'.

**Q.11 · Answer (B) disgoise**
The correct spelling is 'disguise'.

**Q.12 · Answer (D) happiness**
'Brought' needs a noun as its object: 'brought happiness to the face of every citizen'.

**Q.13 · Answer (A) was born in**
One is born 'in' a place: 'was born in Southampton'.

**Q.14 · Answer (B) head over heels**
To fall 'head over heels' in love is to fall deeply in love.

**Q.15 · Answer (B) To avoid getting to the point of an issue**
To 'beat around the bush' is to avoid coming to the point.

**Q.16 · Answer (B) vibrant**
The opposite of 'lifeless' is 'vibrant', full of life and colour; 'muted' is close to lifeless.

**Q.17 · Answer (C) Enrage**
To 'pacify' is to calm someone; its opposite is 'enrage'.

**Q.18 · Answer (B) The food company will be promoting newly launched products at affordable prices.**
'Will be promoting' (the future continuous) is the only future form offered; the others are the present continuous, the simple present and the past perfect.

**Q.19 · Answer (A) bravely**
The blank needs an adverb of manner saying how they trekked: 'bravely'.

**Q.20 · Answer (A) Hasten**
To 'delay' is to put off or slow down; its opposite is 'hasten'.

**Q.21 · Answer (A) surprise**
For a society 'fixated on numbers' it is 'no surprise', only to be expected, that we score our weight with them.

**Q.22 · Answer (D) popular**
BMI is the 'most popular' of these measurements, the one most widely used.

**Q.23 · Answer (D) classifies**
BMI 'classifies' bodies into the categories that follow: underweight, normal, overweight or obese.

**Q.24 · Answer (A) screening**
A useful tool for weight and health 'screening': checking people for possible problems.

**Q.25 · Answer (A) measure**
It should not be 'the single measure', the only yardstick, of what a healthy weight is.

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